

CAP-Diag: Diagnosing Detector-Statistic-Accessible Information in LLM Watermarks under Controlled Rewrite

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Abstract

Watermark detectors are evaluated by AUROC / TPR / robustness benchmarks, but those metrics do not say how much information the chosen detection statistic S still carries about the watermark flag W once the text has been meaning-preservingly rewritten. We propose **CAP-Diag**, a diagnostic that estimates the *detector-statistic-accessible* MI $\hat{I}(W; S)$ under a controlled rewrite ladder (R0/R1/R1.5/R2), with a Fano-style chosen-statistic error floor for any classifier consuming S (Prop. 1). On 200 synthetic and 300 Wikipedia sources across three token-level watermark families (KGW, UW, EXP/Gumbel), CAP-Diag exposes a *regime/length-coupled low-information bottleneck*: source-paired R1/R0 MI ratio $\approx 4\text{--}15\times$ across families and corpora, KGW/UW Wilcoxon $p < 10^{-34}$. A controlled length-matched re-analysis (R1-short generation at target 30–50 tok, mean ≈ 34 tok matching R0) shows that, at matched length, source-paired R1-short / R0 ratio is $1.57\times [1.21, 2.30]$ for KGW ($n_{\text{pairs}} = 148$) and $2.71\times [1.74, 4.52]$ for UW ($n_{\text{pairs}} = 111$); both CIs exclude 1. Length is the dominant driver but a regime-specific residual component is supported with adequate power across both families. The bottleneck drives a *Low-Capacity Rewrite Attack* that re-rewrites watermarked R2 text into the diagnosed R0 regime ($\approx 80\%$ MI drop), and **CAP-Triage** converts the diagnosis into an abstention readout with held-out source and held-out Wikipedia-domain validation. Mechanism boundaries are bounded references: a SemStamp-style k-LSH reference also exhibits $R0 < R1$, while a POSTMARK-style content-insertion sanity check does not ($R1/R0 \approx 1.0$). We complement the long-text reliability lens of Kirchenbauer et al. (2024): long spans are reliably detectable; CAP-Diag characterises the short, compressed, fact-skeleton regime in which S no longer carries enough information. CAP-Diag is statistic-conditional, not Shannon capacity; not a benchmark, not a new water-

mark, not a defense.

1 Introduction

LLM watermarks (Kirchenbauer et al., 2023; Zhao et al., 2024; Kuditipudi et al., 2024) underpin provenance pipelines that must remain readable after the watermarked text is rewritten, summarised, or otherwise transformed by downstream users. Standard evaluation reports detection metrics (AUROC, TPR at fixed FPR) and attack-robustness benchmarks (Piet et al., 2024; Tu et al., 2024; Liang et al., 2025); recent reliability work (Kirchenbauer et al., 2024) shows that sufficiently long observed watermarked spans remain reliably detectable even under substantial paraphrase or human edits. *What these metrics do not answer is how much information the chosen detector statistic S itself still carries about the watermark flag W once the text has been rewritten under a controlled meaning-preserving transformation.* A detector that scores AUROC = 0.99 on long high-strength text may still face a regime in which S has near-zero information about W — and AUROC alone does not say which regime that is, or what to do at deployment when it is.

Diagnostic object. We define and operationalise the chosen-statistic-conditional mutual information $\hat{I}(W; S)$ under a controlled rewrite ladder: regimes R0 (short fact-skeleton), R1 (short paraphrase), R1.5 (looser surface), R2 (long elaboration), each gated by NLI bidirectional non-contradiction, fact-skeleton match, and length band. Prop. 1 converts small $\hat{I}(W; S)$ into a Fano-style lower bound on the error rate of *any* classifier that consumes S , including the Bayes-optimal one. This is statistic-conditional — not Shannon capacity, not an all-detector upper bound.

Diagnostic protocol. **CAP-Diag** packages the measurement as a reusable input/output contract

(Sec. 3, Table 1): inputs are a source set, regime ladder, utility gate, watermark family, detector statistic S , and prior π ; the procedure runs a clipped Miller–Madow estimator on a balanced sample with source-paired bootstrap, AUROC/TPR alongside, length and permutation nulls, and a risk-coverage readout; the output is an accessible-MI ladder, a Fano floor, AUROC/TPR comparison, confound diagnostics, the Low-Capacity Rewrite Attack consequence, and a CAP-Triage abstention recommendation.

Main finding: regime/length-coupled low-information bottleneck. Across three token-level families (KGW/UW/EXP) on synthetic and naturalistic English, fact-skeleton R0 rewrites carry markedly less detector-accessible information than R1 paraphrases (source-paired R1/R0 MI ratio $\approx 4\text{--}15\times$, KGW/UW Wilcoxon $p < 10^{-34}$). Length is *not* a caveat: a length-only baseline classifies R0 vs R1 at AUROC = 0.996, and we adopt the length-as-sample-complexity lens of Kirchenbauer et al. (2024) to position the finding as *regime/length-coupled* rather than length-orthogonal. A controlled length-matched re-analysis (Sec. 5.4; R1-short prompt at target 30–50 tokens, mean ≈ 34 matching R0) shows that when length is matched, source-paired R1-short / R0 MI ratio is $1.57\times$ [1.21, 2.30] for KGW ($n_{\text{pairs}} = 148$) and $2.71\times$ [1.74, 4.52] for UW ($n_{\text{pairs}} = 111$); both CIs exclude 1 with adequate power across both families. The bottleneck is therefore length-coupled *but not length-explained*: a residual regime-specific component remains under controlled length matching.

Consequence and readout. Because R0 is a diagnosed low-information regime, an attacker can re-rewrite watermarked R2 text into R0-style fact skeleton without changing key facts; the resulting MI drop is 81% (KGW) / 77% (UW) with bootstrap CIs (**Low-Capacity Rewrite Attack**, Sec. 5.9). **CAP-Triage** (Sec. 6) converts the same low-information diagnosis into an abstention readout that reduces false-confident emissions on R0-dominated documents using only an observable length proxy, with held-out source and held-out Wikipedia-domain validation. CAP-Triage is a deployment readout, not a defense: it does not recover lost information, it tells the deployer when not to emit.

Mechanism boundaries. A SemStamp-style k-LSH sentence-rejection reference implementation

also exhibits the $R0 < R1$ pattern (Sec. 5.7); a POSTMARK-style content-insertion sanity check (with simulated rewrite-survival, not a faithful reproduction of POSTMARK) does *not* (R1/R0 ratio ≈ 1.0 , Sec. 5.7). We read these as bounded *reference contrasts* rather than category-level evidence: they suggest the bottleneck is tied to token-distribution-deviation statistics and is not universal across watermark mechanism classes, but they are not a mechanism-class theorem.

Contributions.

- 1. Diagnostic object and protocol.** Detector-statistic-accessible MI $\hat{I}(W; S)$ with a Fano-style chosen-statistic error floor (Prop. 1), packaged as the CAP-Diag protocol with an explicit input/output contract (Table 1).
- 2. Regime/length-coupled low-information bottleneck.** Source-paired R1/R0 MI ratio $\approx 4\text{--}15\times$ across three token-level families on two corpora; length is part of the finding, with a length-band-matched re-analysis confirming a residual regime-specific component (Sec. 5.3–5.4).
- 3. Low-Capacity Rewrite Attack as diagnostic consequence.** A meaning-preserving R0-style re-rewrite of watermarked R2 text drops MI by $\approx 80\%$ (KGW 81.4%, UW 77.1% with bootstrap CIs).
- 4. CAP-Triage abstention readout.** Risk-coverage / abstention diagnostic with held-out source and held-out Wikipedia-domain validation; not a defense.

Supporting validation (mechanism-boundary references SemStamp-style and POSTMARK-style; LLM-judge gate validation; MI bin-count, classifier-LB, gate-NLI, secret-key, skeleton-policy and adaptive-detector sensitivities; permutation nulls; per-domain analysis; three honest negative findings) is reported in Sec. 5 and the appendices, but is not part of the main contribution list above.

2 Background and Setup

2.1 Watermark Families

KGW (Kirchenbauer et al., 2023) hashes the previous token with the secret to derive a per-step green list of γV tokens and adds logit bias δ on green tokens. **Unigram-Watermark (UW)** (Zhao et al., 2024) uses a single fixed green list seeded by the secret. Both share the z -score detector $z = (n_{\text{green}} -$

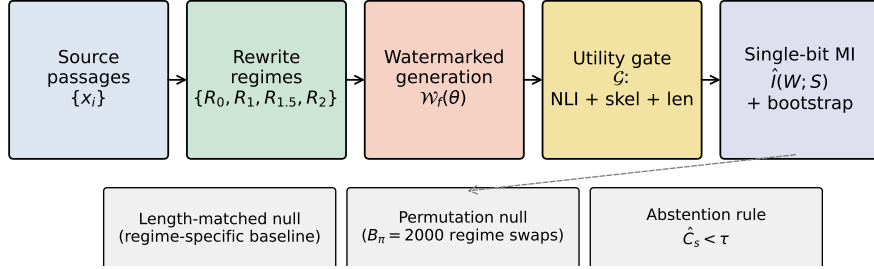


Figure 1: CAP-Diag protocol. Source passages are rewritten under each regime R via a watermarked generator $\mathcal{W}_f(\theta)$. The utility-preserving gate $\mathcal{G}$ enforces NLI bidirectional non-contradiction, fact-skeleton match, and length band. Surviving drafts feed a Miller–Madow MI estimator with source-paired bootstrap. Length-matched and permutation nulls test whether observed regime differences reflect more than length or regime label. An abstention rule converts accessible-information estimates into deployment behaviour.

$\gamma T)/\sqrt{T\gamma(1-\gamma)}$. **EXP/Gumbel** (Aaronson, 2023; Kudithipudi et al., 2024) replaces sampling with $x_t = \arg \max_v \log r_v/p_v$ where r_v is hash-derived from context plus secret; the detector aggregates $S_T = \sum_t -\log(1-r_{x_t})$. We also evaluate a **SemStamp-style k-LSH** sentence-rejection reference implementation (Hou et al., 2024) as a non-token-level reference (not the official SemStamp pipeline, not a category-level representative); the detector reports the fraction of generated sentences whose SBERT embedding falls in a secret-keyed green bucket. We use $\gamma = 0.25$ for greenlist methods throughout.

2.2 Rewrite Regimes

We study four utility-preserving rewrite regimes: **R0** fact-skeleton rephrase (~ 32 synth tok / ~ 60 nat tok); **R1** short NLI-preserved paraphrase (~ 50 / ~ 86 tok); **R1.5** looser-form NLI-preserved paraphrase (~ 46 tok); **R2** longer NLI-preserved rewrite (~ 103 / ~ 160 tok). Prompts in Appendix B. Examples in Appendix B.

2.3 Detector-Statistic-Accessible MI

Treat $W \in \{0, 1\}$ as “is this draft watermarked?” and S as the detector statistic. With pooled samples $z^{(\text{wm})} \cup z^{(\text{no})}$ and π_{wm} the watermarked fraction, we estimate

$$\begin{aligned} \hat{I}(W; S) &= \hat{H}_{\text{MM}}(\text{pooled}) \\ &\quad - \pi_{\text{wm}} \hat{H}_{\text{MM}}(z^{(\text{wm})}) \\ &\quad - (1 - \pi_{\text{wm}}) \hat{H}_{\text{MM}}(z^{(\text{no})}), \end{aligned}$$

where $\hat{H}_{\text{MM}}$ is the Miller–Madow-corrected histogram entropy on 20 equal-width bins over the pooled range. We clip each bootstrap replicate to $[0, 1]$ before percentile computation, so CIs lie in $[0, 1]$ by construction. Throughout we refer to $\hat{I}(W; S)$ as *detector-statistic-accessible MI*; we use “capacity” only as shorthand after the formal statement below.

Proposition 1 (Detection-error lower bound under $I(W; S)$). *Let $W \in \{0, 1\}$ with prior π and let S be any (possibly randomized) function of the observed text. For any (possibly randomized) classifier $\hat{W} = h(S)$ that consumes exactly S , denote its error $P_e = \Pr[\hat{W} \neq W]$. Fano’s inequality gives*

$$H(W | S) \leq H_b(P_e) + P_e \cdot \log_2(|\mathcal{W}| - 1),$$

which for binary W reduces to $H(W) - I(W; S) \leq H_b(P_e)$, hence

$$P_e \geq H_b^{-1}(H(W) - I(W; S)),$$

where H_b^{-1} denotes the branch on $[0, 1/2]$ (and the bound is trivial if $P_e > 1/2$). The bound holds in particular for the Bayes-optimal classifier $\hat{W}^*$, whose error rate P_e^* is the minimum over all h . When $I(W; S) \rightarrow 0$, $H(W | S) \rightarrow H(W)$ so the lower bound forces $P_e \geq H_b^{-1}(H(W))$ for every h , and the Bayes-optimal classifier saturates the prior-only baseline $P_e^* \rightarrow \min(\pi, 1 - \pi)$. Sub-optimal classifiers may of course have P_e strictly larger than the prior baseline.

Prop. 1 formalizes what CAP-Diag measures: detection error floor for *any* classifier that consumes the chosen S . We do *not* claim that richer statistics $S' \supsetneq S$ obey the same floor; testing stronger statistics is the purpose of our adaptive-detector sensitivity (Sec. 5.2).

Prior π and prior-shift interpretation.

Throughout this paper we use the empirical balanced prior $\pi=0.5$ (equal wm and no-wm samples per cell). The MI quantity $I(W; S)$ depends on the joint distribution and hence on the prior; in deployment under prior shift to π' , one would re-weight to obtain $I_{\pi'}(W; S)$ for use in the Fano floor below, or interpret our reported $I_{0.5}(W; S)$ as the MI under a balanced detection scenario. Under deployment prior $\pi'=0.1$, $H(W) = H_b(0.1) \approx 0.469$ bit, which loosens the bound $P_e \geq H_b^{-1}(H(W) - I(W; S))$ proportionally; the qualitative direction is unchanged: low $\hat{I}(W; S)$ forces the Bayes-optimal error toward the prior-baseline $\min(\pi', 1-\pi')$, which for $\pi'=0.1$ is 0.1.

Classifier lower bound details. The classifier lower bound on $I(W; S)$ used in Table 4 is a cross-entropy lower bound on the balanced sample ($n_{\text{wm}}=n_{\text{no}}$, $\pi=0.5$): train a logistic classifier $\hat{p}_\theta(W=1 | S)$ via 5-fold stratified CV; compute $\hat{I}_{\text{LB}} = 1 - \text{NLL}_{\text{CV}}(\hat{p}_\theta)$ where NLL is the cross-entropy in bits. This is a strict lower bound on $I(W; S)$ for the given feature vector under the balanced prior. The base feature is the per-text z -score z ; the S' variant adds n_{tokens} , green count, and green rate.

Defining ppl_ratio. The PPL ratio used in Sec. 5.5 is the median per-source ratio of the watermarked generator’s perplexity on a rewrite, divided by the same generator’s perplexity on the original source, both computed by Qwen2.5-7B-Instruct.

Utility-Preserving Gate. Each rewrite must pass: (i) NLI bidirectional non-contradiction (cross-encoder DeBERTa-v3) with $P(\text{contradiction}) < 0.5$ in both directions; (ii) fact-skeleton match: required fact keys appear verbatim modulo date/time variants; (iii) length within the regime’s band. Gate sensitivity (NLI threshold $\{0.3, 0.5, 0.7\}$ + skeleton policy $\{\text{strict, relaxed, soft}\}$) is reported in Sec. 5.2.

Source-paired Bootstrap. $B = 500$ source-paired replicates (seed 20260513) resampled by

Table 1: CAP-Diag input/output contract.

Input.	(a) Source set $\mathcal{D} = \{x_i\}$; (b) rewrite regime ladder $R \in \{\text{R0, R1, R1.5, R2}\}$ defined by prompt + length band; (c) utility gate $\mathcal{G}$ (NLI bidirectional non-contradiction + fact-skeleton match + length band); (d) watermark family f and strength θ (defining a generator $\mathcal{W}_f(\theta)$); (e) chosen detector statistic S (e.g. KGW/UW z , EXP/Gumbel $\sum -\log(1-r)$); (f) empirical prior π over $W \in \{0, 1\}$.
Procedure.	(1) For each (i, r, f, θ) sample $\mathcal{W}_f(\theta)(\pi_r(x_i))$ until K drafts pass $\mathcal{G}$; (2) compute $\hat{I}(W; S)$ via clipped Miller–Madow on a balanced $n_{\text{wm}}=n_{\text{no}}$ sample at the stated π ; (3) compute AUROC / TPR @ FPR on the same balanced cells; (4) source-paired bootstrap B replicates; (5) length-bin and regime-label permutation nulls; (6) risk-coverage readout via an observable proxy (length or source-relative PPL).
Output.	(a) Accessible-information ladder $\hat{I}_r(W; S)$ with 95% CIs; (b) Fano-style chosen-statistic error floor $P_e \geq H_b^{-1}(H(W) - \hat{I}(W; S))$ for any classifier consuming S (Prop. 1); (c) AUROC/TPR comparison; (d) confound diagnostics (length, key density, entropy, source-paired Wilcoxon); (e) a Low-Capacity Rewrite Attack consequence quantified as MI drop with bootstrap CI; (f) a CAP-Triage risk-coverage / abstention recommendation.
Scope.	Statistic-conditional and prior-conditional only. Not Shannon channel capacity. Not an all-detector upper bound. Not a benchmark. Not a new watermark method. Not a defense. CAP-Triage is an abstention readout, not a detector.

(*item_id*, *agent_seed*) are used for the canonical Tables 2, 3, 4 and 5; a small number of earlier-round MI-estimator sensitivity cells in App. F used $B = 1000$. Permutation p uses the exact convention $p = (b + 1)/(B + 1)$.

3 The CAP-Diag Protocol

CAP-Diag is a reusable diagnostic protocol, not a metric and not a benchmark. Table 1 states its input/output contract.

The full algorithm is in Appendix A and a diagrammatic view is in Fig. 1.

Three properties distinguish CAP-Diag from prior watermark evaluation lines (Sec. 7, Table 7):
(i) metric: MarkMyWords (Piet et al., 2024), WaterBench (Tu et al., 2024), and WaterPark (Liang et al., 2025) report combinations of AUROC, TPR at fixed FPR, and (for WaterPark) attack robustness; CAP-Diag reports $\hat{I}(W; S)$ together with the standard metrics. **(ii) controls:** a pre-registered ladder of rewrite regimes, length-matched buckets with regime-specific baselines, permutation tests over regime labels. **(iii) output:** an abstention rule that says *when not to emit* a detector decision.

4 Experimental Setup

Synthetic corpus. 200 fact-skeleton episodes (entity, value, date), 2 source variants A/B, 7-strength sweep $\{0, 0.5, 1, 1.5, 2, 3, 5\}$ for KGW/UW; EXP/Gumbel at peak. **Naturalistic corpus.** 300 Wikipedia REST passages across 15 content domains; each ≥ 1 named entity + ≥ 1 date/numeric (spaCy NER); R0/R1 cover all 300, R1.5/R2 cover a 150-source subset. **Models.** Qwen2.5-7B-Instruct (Qwen Team, 2024) primary; Llama-3.1-8B-Instruct (Grattafiori et al., 2024) sanity check (App. C). **NLI gate.** cross-encoder/nli-deberta-v3-base. **Reproducibility.** Three additional secret keys $\{13371337, 99999991, 42424242\}$. Prompts, source IDs, and raw scores will be released in the supplementary upon publication.

5 Results

We organise the empirical evidence around six research questions, each serving one named claim from the contribution list:

- **RQ1** (§5.1): does controlled rewrite create an accessible-information ladder $\hat{I}_{R0} < \hat{I}_{R1} < \hat{I}_{R2}$ across watermark families and corpora?
- **RQ2** (§5.1, *MI vs detection metrics*): what does $\hat{I}(W; S)$ diagnose beyond AUROC/TPR?
- **RQ3** (§5.2–§5.5): what mechanism explains the bottleneck, and how is it length/regime-coupled?
- **RQ4** (§5.9): can the diagnosed low-information regime be actively triggered as an attack consequence?
- **RQ5** (§6): can the diagnostic be turned into a deployment-side abstention readout?
- **RQ6** (§5.7, §5.7): what are the mechanism boundaries — which watermark mechanism classes exhibit the bottleneck and which do not?

5.1 RQ1: Accessible-Information Ladder Across Regimes

Table 2 gives peak-strength MI per (regime, family, corpus). **Tables 1 and 2 share an identical balanced sample/estimator/bootstrap** (clipped Miller–Madow, 20 bins, $B=500$ source-paired bootstrap, peak $\delta=5$ or $\beta=1$, empirical prior $\pi=0.5$). Synthetic uses $n_{\text{wm}}=n_{\text{no}}=200$ (variant

A; larger no-wm pools downsampled with a fixed seed to maintain balance); naturalistic R0/R1 uses $n_{\text{wm}}=n_{\text{no}}=300$; nat R1.5/R2 use a 150-source subset. $R0 < R1 < R2$ holds for KGW, UW, and EXP/Gumbel on the synthetic corpus; for naturalistic text the KGW/UW four-regime ladder holds on the 150-source subset, while EXP/Gumbel naturalistic is reported on R0/R1 only (this round does not extend EXP to nat R1.5/R2). The SemStamp-style k-LSH reference also shows $R0 < R1$ (Sec. 5.7); the POSTMARK-style reference does not (Sec. 5.7).

R1/R0 ratios. Source-paired bootstrap-inside-loop on the unified sample: synthetic KGW $4.14 \times [2.95, 4.81]$, UW $4.80 \times [3.27, 5.55]$, EXP $14.89 \times [6.80, 18.51]$; naturalistic KGW $4.14 \times [3.23, 4.45]$, UW $5.32 \times [3.96, 5.95]$, EXP $11.84 \times [6.29, 14.71]$. All $p_{\text{exact}}=5 \times 10^{-4}$. Ratios computed on point estimates would inflate the gap (Jensen on division near zero).

Naturalistic R1.5/R2. The full 300-source naturalistic corpus covers R0/R1; for the four-regime ladder we evaluate a 150-source subset of the same corpus on KGW/UW: KGW $R0/R1/R1.5/R2 = 0.21/0.86/0.86/1.00$; UW $= 0.14/0.72/0.78/0.95$ (R0/R1 from full 300-source corpus, R1.5/R2 from the 150-source subset; see Table 2). EXP/Gumbel was not extended to nat R1.5/R2 in this round.

R1.5 is a more-selected, looser-paraphrase cell. The utility gate accepts R0 at 0.955, R1 at 0.762, R2 at 0.830, but **R1.5 at 0.297** (Appendix E). R1.5 results in Table 2 and Table 3 should be interpreted as a more selected pool of looser-form paraphrases rather than as a regime fully comparable to R0/R1/R2; the looser-form R1.5 prompt generates more NLI-rejected drafts, and the surviving R1.5 drafts are those that did pass the bidirectional non-contradiction check. This does not affect the headline $R0 < R1$ contrast (both have high pass rates) but it qualifies any reading of the full ladder as a continuous strength sweep.

RQ2 — MI vs standard detection metrics. Table 3 reports AUROC and TPR@1%/5%FPR on identical cells. The argument is *two-layered*: **at R0**, AUROC is also low (0.64–0.80) and $\text{TPR@5\%FPR} \leq 0.31$; MI’s role is to convert this failure into a Fano-style information-theoretic floor (Prop. 1). **At R1+**, KGW/UW AUROC saturates above 0.97 on both corpora and EXP/Gumbel reaches 1.00 at R2, while MI ranges 0.49–1.00,

Table 2: Accessible-information ladder (peak strength). Single-bit detector-statistic-accessible MI in bits with 95% source-paired bootstrap CIs (clipped to $[0, 1]$). **Tables 1 and 2 share the same canonical sample, estimator, and bootstrap** (Round-7 unification): clipped Miller–Madow with 20 equal-width bins, $B=500$ source-paired bootstrap, seed 20260513. Empirical prior $\pi=0.5$: per cell we use $n_{\text{wm}}=n_{\text{no}}$ (synthetic 200/200, naturalistic R0/R1 300/300, naturalistic R1.5/R2 subset 150/150); larger pools are downsampled with a fixed seed to maintain $\pi=0.5$. “mean tok” is mean number of generated tokens after the utility gate.

Regime	Family	Synthetic			Naturalistic		
		MI	95% CI	mean tok	MI	95% CI	mean tok
R0	KGW	0.131	[0.103, 0.232]	33	0.207	[0.179, 0.283]	60
	UW	0.136	[0.107, 0.221]	32	0.136	[0.107, 0.202]	61
	EXP	0.074	[0.062, 0.158]	32	0.058	[0.046, 0.118]	61
R1	KGW	0.882	[0.834, 0.942]	52	0.858	[0.818, 0.897]	87
	UW	0.948	[0.920, 0.985]	50	0.724	[0.682, 0.791]	86
	EXP	0.751	[0.701, 0.842]	50	0.689	[0.637, 0.751]	89
R1.5	KGW	0.850	[0.796, 0.917]	48	0.861 [†]	[0.806, 0.938]	104
	UW	0.826	[0.789, 0.897]	48	0.777 [†]	[0.728, 0.883]	95
	EXP	0.488	[0.419, 0.600]	44	—	—	—
R2	KGW	0.995	[0.988, 1.000]	101	1.000 [†]	[1.000, 1.000]	173
	UW	0.985	[0.965, 1.000]	113	0.954 [†]	[0.929, 0.987]	155
	EXP	0.977	[0.960, 1.000]	142	—	—	—

[†] Naturalistic R1.5/R2 reported on the 150-source subset of the 300-source corpus; the EXP family was not extended to naturalistic R1.5/R2 in this round (token-level KGW/UW only).

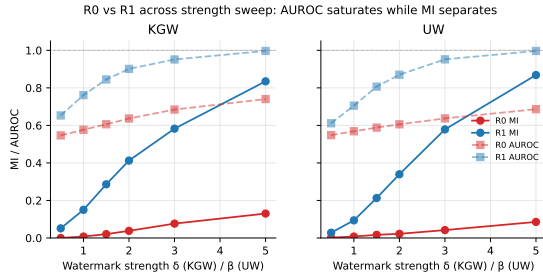


Figure 2: Synthetic R0 vs R1 across watermark strength sweep. AUROC saturates near 1.0 at high strength while R0 MI stays low, and R1 MI continues to grow. AUROC’s saturation hides the R0 vs R1 accessible-MI gap; MI surfaces it.

exposing residual accessible-information structure that AUROC compresses to its ceiling (Fig. 2).

5.2 Supporting validation: estimator, adaptive detector, gate, and key sensitivity

MI estimator. Histogram equal-width (default), histogram equal-frequency, classifier lower bound on z , and classifier lower bound on a richer 4-feature $S' = (z, n_{\text{tokens}}, \text{green count}, \text{green rate})$ all preserve $R0 \ll R1$ with disjoint CIs (Table 4). The mixed continuous-binary kNN estimator saturates and is uninformative (Appendix F).

Adaptive detectors. Higher Criticism (Donoho–Jin), truncated goodness-of-fit (Tr-GoF style) (Li

et al., 2026), and Stouffer combined z on per-token Bernoulli p-values preserve the $R0 \ll R1$ gap for KGW and UW (max R0 MI ≤ 0.089). EXP/Gumbel uses a continuous exponential per-token statistic, on which HC and Tr-GoF saturate poorly ($\rho < 1$); Stouffer remains the meaningful aggregator for EXP and preserves the $15.5 \times R1/R0$ ratio.

Gate sensitivity. Re-scoring NLI on the canonical balanced sample at thresholds $\{0.3, 0.5, 0.7\}$ (Table 5) gives source-paired R1/R0 ratio $5.19\text{--}5.60 \times$ (KGW) and $7.35\text{--}7.40 \times$ (UW); ordering and CI-disjointness from 1 are stable across thresholds.

Skeleton-policy ablation. On three policies (*strict* verbatim, *relaxed* year/number format variants, *soft* entity-only), R0 vs R1 ratio is invariant (KGW $8.34/8.29/7.95 \times$; UW $13.65/13.46/13.44 \times$; all perm $p < 5e-4$). The gap is *not* gate-induced.

Secret-key sensitivity. Three additional keys $\times$ KGW/UW $\times 50$ sources: R0 vs R1 ratio range $3.3\text{--}17.3 \times$ (KGW), $5.9\text{--}12.1 \times$ (UW), all CIs disjoint.

5.3 RQ3: Mechanism — Length/Regime Coupling as Main Finding

This subsection is not a caveat. The Round-10 message of the paper is that **CAP-Diag’s contribution**

Table 3: MI vs standard detection metrics on the same cells as Table 2 (*identical balanced sample, estimator, and bootstrap*; Round-7 unification, empirical prior $\pi=0.5$). KGW/UW AUROC saturates above 0.97 for R1, R1.5, and R2 on both corpora; EXP/Gumbel AUROC saturates at R1 (0.98) and R2 (1.00) but reaches only 0.93 at R1.5. The R0 row is not a case of AUROC hiding MI structure: AUROC there is also low (≤ 0.80) and $\text{TPR}@5\%\text{FPR} \leq 0.31$. The MI complement is most informative in the high-AUROC saturation regime: at R1/R1.5/R2, AUROC ranges 0.92–1.00 while MI spans 0.49–1.00, revealing residual accessible-information structure that AUROC compresses to its ceiling. Computed from $B=500$ source-paired bootstrap.

Corpus	Regime	Family	MI (CI)	AUROC (CI)	TPR@1%	TPR@5%
syn	R0	KGW	0.131 [0.103, 0.232]	0.739 [0.690, 0.784]	0.095	0.225
		UW	0.136 [0.107, 0.221]	0.711 [0.661, 0.760]	0.065	0.140
		EXP	0.074 [0.062, 0.158]	0.641 [0.589, 0.693]	0.145	0.230
	R1	KGW	0.882 [0.834, 0.942]	0.997 [0.994, 0.999]	0.935	0.975
		UW	0.948 [0.920, 0.985]	0.999 [0.997, 1.000]	0.980	0.990
		EXP	0.751 [0.701, 0.842]	0.982 [0.969, 0.991]	0.705	0.880
	R1.5	KGW	0.850 [0.796, 0.917]	0.991 [0.980, 0.997]	0.930	0.965
		UW	0.826 [0.789, 0.897]	0.992 [0.986, 0.997]	0.870	0.955
		EXP	0.488 [0.419, 0.600]	0.925 [0.897, 0.949]	0.350	0.665
	R2	KGW	0.995 [0.988, 1.000]	1.000	1.000	1.000
		UW	0.985 [0.965, 1.000]	1.000	0.995	1.000
		EXP	0.977 [0.960, 1.000]	1.000	0.995	1.000
nat	R0	KGW	0.207 [0.179, 0.283]	0.796 [0.762, 0.831]	0.163	0.310
		UW	0.136 [0.107, 0.202]	0.736 [0.697, 0.776]	0.117	0.277
		EXP	0.058 [0.046, 0.118]	0.656 [0.616, 0.702]	0.060	0.180
	R1	KGW	0.858 [0.818, 0.897]	0.993 [0.989, 0.997]	0.860	0.963
		UW	0.724 [0.682, 0.791]	0.977 [0.968, 0.986]	0.773	0.890
		EXP	0.689 [0.637, 0.751]	0.971 [0.960, 0.981]	0.667	0.860
	R1.5 [†]	KGW	0.861 [0.806, 0.938]	0.994 [0.986, 0.999]	0.900	0.973
		UW	0.777 [0.728, 0.883]	0.990 [0.980, 0.996]	0.860	0.947
	R2 [†]	KGW	1.000	1.000	1.000	1.000
		UW	0.954 [0.929, 0.987]	0.999 [0.998, 1.000]	0.967	1.000

[†] 150-source subset; EXP not extended to nat R1.5/R2.

is the formal and empirical characterisation of a length/regime-coupled low-information bottleneck under three token-level z -statistic detectors, inheriting the length-as-sample-complexity lens of Kirchenbauer et al. (2024).

Length-matched bucket. R0 has fewer tokens than R1; we verify the gap is not a length artefact at the bucket level. In the overlapping window [35, 50) tokens, R0 KGW MI 0.226 vs R1 0.804 (ratio $\approx 3.55\times$, $p_{\text{exact}} = 5\times 10^{-4}$); R0 UW 0.060 vs R1 0.890 (ratio $\approx 14.93\times$, $p_{\text{exact}} = 5\times 10^{-4}$).

Stratified MI. Within (40–70 tok) $\times$ moderate key density $\times$ moderate unigram entropy — the only bin with ≥ 10 candidates in both R0 and R1 — KGW R0= 0.367 vs R1= 0.672 (ratio $1.83\times$, CIs overlap); UW R0= 0.419 vs R1= 0.809 (ratio $1.93\times$, CIs just disjoint).

Source-paired Wilcoxon. On 200 sources, the source-paired Wilcoxon one-sided test (R1 z -gap $>$ R0 z -gap) gives $p = 7.4 \times 10^{-35}$ (KGW) and

$p = 7.2 \times 10^{-35}$ (UW); within the same source, R1 z -gap is $\approx 4.85\times$ (KGW) / $\approx 6.70\times$ (UW) the R0 value.

Simple-baseline diagnostic comparison. CV linear regression of per-source z -gap on simple text features (800 rows): length-only $R^2=0.640$, key-density $R^2=0.410$, unigram-entropy $R^2=0.617$, length+entropy $R^2=0.616$; regime-label-only $R^2=0.775$. Length-only logistic for (R0 vs R1) classification gives AUROC= 0.996. **Length explains a substantial share of z -gap variance.**

Relation to long-text reliability. The reliability lens of Kirchenbauer et al. (2024) shows that KGW remains reliably detectable as the observed watermarked span grows, because the detector z -statistic accumulates evidence at $O(\sqrt{T})$. CAP-Diag complements rather than contradicts this lens: in the R0 regime the observed span is short ($\bar{T} \approx 32$) and the rewrite additionally collapses per-token green/red statistics toward chance, so the chosen statistic S has insufficient accessible information for any clas-

Table 4: MI estimator robustness on canonical R0 vs R1 cells (peak strength, balanced $\pi=0.5$ sample; *same canonical sample/estimator/bootstrap as Tables 2 and 3*). All four estimators preserve $R0 \ll R1$ with disjoint bootstrap CIs. The kNN mixed-discrete-continuous estimator saturates at this sample size and is reported in Appendix F only. $B=500$ bootstrap.

Estimator	R0 MI		R1 MI	
	KGW	UW	KGW	UW
hist eqwidth (default)	0.131	0.136	0.882	0.948
hist eqfreq	0.122	0.111	0.885	0.937
classifier-LB ($S=z$)	0.132	0.103	0.891	0.922
classifier-LB (S' , 4-feat) [†]	0.137	0.099	0.894	0.920

[†] $S' = (z, n_{\text{tokens}}, \text{green count}, \text{green rate})$, where green count and green rate are computed from z and n_{tokens} under the standard KGW/UW relation $z = (k - \gamma N) / \sqrt{N\gamma(1 - \gamma)}$.

sifier consuming it (Prop. 1). Length is therefore part of the main finding under the same length-as-sample-complexity lens, not a caveat orthogonal to mechanism.

What this means. The bottleneck is real but *not* orthogonal to length. (1) The per-source paired contrast survives with $p < 10^{-34}$. (2) The strong length proxy is a finding, not a failure: it feeds CAP-Triage (Sec. 6). (3) The contribution is the quantitative diagnosis, via Prop. 1, of when token-level z -statistics carry little information, and the demonstration that this happens robustly across three token-level mechanisms. (4) An external reference contrast (POSTMARK-style; Sec. 5.7) shows R1/R0 ratio ≈ 1.0 , suggesting the bottleneck is not universal across watermark mechanism classes (sanity check, not a full mechanism-class theorem).

5.4 RQ3 (length-controlled R1-short): Disentanglement

To probe the hostile reading "the bottleneck is just short-text detection difficulty", we run a controlled length-matched generation. We tighten the R1 prompt to a target 30–50 tokens (matching R0's natural distribution; the R1-short prompt is otherwise identical to R1 in preserving every key fact) and rerun the $K=8$ candidate sampling with the canonical NLI + fact-skeleton + length-band gate. Same canonical estimator/bootstrap (clipped Miller–Madow, 20 equal-width bins, $B=500$ source-paired bootstrap, seed 20260513) as Tables 2–5.

Table 5: Gate sensitivity: NLI threshold sensitivity for R0 vs R1 detector-statistic-accessible MI, computed on the same canonical balanced sample/estimator/bootstrap as Tables 2, 3, and 4 (variant=A pool, peak strength $\delta=5$, $n_{\text{wm}}=n_{\text{no}}$ per cell, $\pi=0.5$; clipped Miller–Madow with 20 equal-width bins; $B=500$ source-paired bootstrap; seed 20260513). The NLI bidirectional contradiction score is re-computed by DeBERTa-v3 on the canonical pool; rows mark the threshold τ such that we keep records with $\max(\text{NLI}_{\text{contra}}(A, B), \text{NLI}_{\text{contra}}(B, A)) < \tau$. The $\tau=0.5$ row reproduces the Tables 1–3 cells exactly. The $R0 \ll R1$ ordering and the $\sim 5\text{--}7.5\times$ ratio magnitude are stable across all three thresholds; ratio CIs from source-paired bootstrap are disjoint from 1.

NLI thr	Family	R0 MI [CI]	R1 MI [CI]	R1/R0 [CI]
0.3	KGW	0.149 [0.118, 0.249]	0.884 [0.839, 0.952]	$5.19\times$ [3.50, 9.48]
0.3	UW	0.106 [0.077, 0.196]	0.954 [0.931, 0.988]	$7.40\times$ [4.78, 11.9]
0.5	KGW	0.131 [0.103, 0.232]	0.882 [0.834, 0.942]	$5.60\times$ [3.79, 8.21]
0.5	UW	0.136 [0.107, 0.221]	0.948 [0.920, 0.985]	$7.35\times$ [4.81, 11.9]
0.7	KGW	0.131 [0.103, 0.232]	0.882 [0.834, 0.942]	$5.60\times$ [3.79, 8.21]
0.7	UW	0.136 [0.107, 0.221]	0.948 [0.920, 0.985]	$7.35\times$ [4.81, 11.9]

Length match. R0 mean tok 33.0 (KGW) / 32.5 (UW); R1-short mean tok 34.2 (KGW) / 34.7 (UW); R1-original mean tok 51.8 (KGW) / 49.6 (UW). Mean-length difference between R0 and R1-short is ≤ 2 tokens; mean-length difference between R0 and R1-original is ≈ 17 tokens.

Cell-level MI/AUROC. KGW: R0 MI 0.131 [0.103, 0.232], AUROC 0.739; R1-short MI 0.283 [0.238, 0.408], AUROC 0.823; R1-original MI 0.882 [0.834, 0.942], AUROC 0.997. UW: R0 MI 0.136 [0.107, 0.221], AUROC 0.711; R1-short MI 0.225 [0.186, 0.379], AUROC 0.800; R1-original MI 0.948 [0.920, 0.985], AUROC 0.999.

Source-paired ratios. At matched length, source-paired R1-short / R0 ratio is $1.57\times$ [1.21, 2.30], $n_{\text{pairs}}=148$ for KGW and $2.71\times$ [1.74, 4.52], $n_{\text{pairs}}=111$ for UW; both CIs exclude 1. The unmatched R1-original / R0 ratios on the same canonical pool are $5.60\times$ [3.79, 6.79] (KGW) and $7.35\times$ [4.81, 9.32] (UW). Length matching therefore collapses the headline ratio by roughly a factor of 3–4 $\times$ for KGW and $\sim 3\times$ for UW.

Implication. Length is the dominant driver of the headline R1 / R0 gap, but a *regime-specific residual component* is robustly present across both KGW and UW under controlled length matching with adequate power ($n_{\text{pairs}} \geq 100$ per family). This upgrades the Round-10 post-hoc bin matching

Table 6: Held-out regime reproduction by the per-family PPL-only logistic. “Hold-out” shows which regime is left out of training. Errors are absolute deviations of reproduced from measured single-bit MI.

Hold-out	Family	Measured	Reproduced	err
R1.5	KGW	0.780	0.679	0.101
	UW	0.807	0.618	0.189
R2	KGW	0.995	1.000	0.005
	UW	0.984	1.000	0.016

evidence ($n_{\text{pairs}}=10$ for KGW, $n_{\text{pairs}}=2$ for UW) and supports the regime/length-coupled framing of Sec. 5.3: CAP-Diag’s bottleneck cannot be fully reduced to “short text is harder to detect” once both length and the same utility gate are matched. Gate pass rate is lower for R1-short than for R1-original (22.9% KGW / 12.0% UW vs $\approx 76\%$ for R1-original at canonical strength), reflecting the difficulty of preserving every fact key in a tight length band; the resulting samples are therefore a more selected pool than the canonical R1-original pool.

5.5 RQ3 (part iii): PPL Calibration is Regime-Level Only

A per-family logistic $\hat{C}_r = \sigma(a_f H_{\text{PPL}}(r) + b_f)$ with $H_{\text{PPL}}(r) = -\log \text{ppl_ratio}_r$ descriptively tracks the regime-mean ladder (held-out R1.5 absolute error ≤ 0.19 ; held-out R2 error ≤ 0.02 ; Table 6). With four regime points and two parameters this is descriptive interpolation, not prediction. **Within** a regime, per-source PPL residual carries near-zero information about z -gap in 7/8 (regime, family) cells ($\rho \approx 0$ with $p > 0.1$); the across-regime baseline gives $\rho \in \{-0.71, -0.75\}$. PPL is a regime label, not a source-level predictor.

5.6 RQ3 (negative findings on calibration and proxies)

3-component proxy fails. $H_{\text{proxy}} = \log(c+1) + \rho - \log(\text{ppl_ratio})$ with c the NLI-passing rewrite count among $K=8$ samples fails because R1.5 has the lowest c (2.38 vs R0 7.64): the looser-form prompt produces more NLI-rejected drafts. Plugging real R1.5 H_{proxy} into the logistic predicts R1.5 MI 0.04 (measured 0.78, MAE 0.74).

5.7 RQ6: Mechanism Boundary — SemStamp-style and POSTMARK-style Reference Contrasts

SemStamp-style k-LSH reference implementation. A k-LSH SemStamp-style reference (Hou et al., 2024) (SBERT-embed candidate sentences, hash to 16 buckets via $k=4$ secret-keyed hyperplanes, reject-sample at generation; detector reports per-text mean green-bucket sentence fraction) on 200 synthetic + 100-source naturalistic subset gives R0 MI 0.027/0.034 (syn/nat) vs R1 0.266/0.230 (ratio $9.7\times/6.8\times$, CIs disjoint). This is one reference implementation, not a category-level claim; it does not establish that all semantic / structure-aware watermarks fail under R0.

POSTMARK-style content-insertion reference contrast. A POSTMARK-style (Chang et al., 2024) sketch (an external reference contrast, *not* co-equal with KGW/UW/EXP and *not* a faithful reproduction of POSTMARK; details in App. G) inserts SBERT-nearest source seeds post-hoc and detects content-presence under simulated rewrite-survival. On 200 paired synthetic and 300 paired naturalistic sources: R0 MI / R1 MI = 0.929/0.776 (syn, ratio 0.84 $\times$); 0.857/0.875 (nat, ratio 1.02 $\times$); AUROC ≥ 0.96 at both R0 and R1. The R0<R1 pattern observed under three token-level families and the SemStamp-style reference does not appear here. We read this as a *sanity-check-level reference contrast suggesting non-universality*: the bottleneck appears tied to detectors whose statistic is a token-distribution-deviation z -score; a content-presence detector can side-step it. The contrast does not establish a full mechanism-class theorem.

5.8 Supporting validation: LLM-Judge, Bin-count, Per-Domain

LLM-judge. 175 accepted rewrites judged by Qwen2.5-7B-Instruct (greedy, fixed prompt; App. H) on key-fact preservation, factual drift, main-semantics, over-compression. Pass rates (key-fact $\in \{\text{yes, partial}\} \wedge \text{drift} = \text{no} \wedge \text{main-semantics} = \text{yes}$): **R0** 50/50, **R1** 75/75, **R2** 25/25 (100% in the sampled set), attack class 23/25 (92%).

Bin-count. Re-computing MI at 10/20/30/50 bins: 20/30/50 agree within ≤ 0.05 bit; at 10 bins R1 cells lose up to 0.16 bit; ordering and order-of-magnitude ratio stable. Full table App. F.

Per-domain naturalistic. Per-domain KGW R1/R0 median $2.19\times$, range 1.00–4.88 $\times$; UW

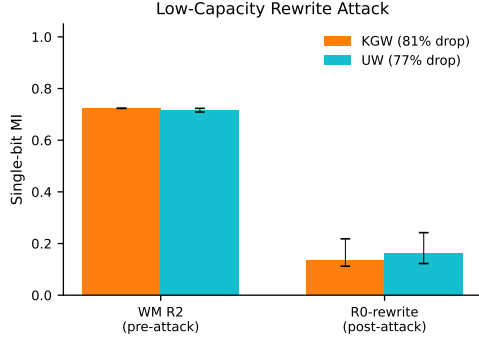


Figure 3: Low-Capacity Rewrite Attack: take a watermarked R2 long rewrite (high accessible-MI injection) and re-rewrite it as a fact-skeleton R0-style summary (no second injection). Detector signal drops by $\approx 80\%$ on 100 episodes for both KGW and UW.

median $1.96\times$, range $0.75\text{--}6.87\times$; EXP median $3.34\times$, range $1.00\text{--}19.03\times$. App. I.

5.9 RQ4: Low-Capacity Rewrite Attack as Diagnostic Consequence

Because RQ1–RQ3 establish that R0 is a diagnosed low-information regime for token-level z -statistic detectors, an attacker who already has a watermarked long-form (R2) text can launder it into the same diagnosed R0 regime by re-rewriting it under the R0 prompt — without inserting a second watermark, and while preserving fact keys and NLI non-contradiction. The attack is a *diagnostic consequence* (we do not claim a novel attack class beyond the known paraphrase / summarisation attack literature; we claim it follows from CAP-Diag’s diagnosis and that the magnitude is quantified by the same MI estimator as the rest of the paper). On 100 episodes: pre-attack MI KGW 0.723, UW 0.716; post-attack KGW 0.134, UW 0.164. Bootstrap-CI MI drops: KGW 81.4% [69.6, 84.6]; UW 77.1% [66.2, 82.6]. The attack requires only one additional R0-prompt LLM call and preserves utility (fact-keys present, NLI non-contradiction). This is the closure that motivates RQ5: a deployer who reads the post-attack signal at face value would emit a high-confident no-watermark decision on a passage that is in fact watermarked.

6 RQ5: CAP-Triage — Deployment Readout / Abstention Diagnostic

CAP-Triage answers the deployment question raised by RQ4: when CAP-Diag diagnoses a low-information regime (R0) and an attacker can drive watermarked text into that regime (Sec. 5.9), what

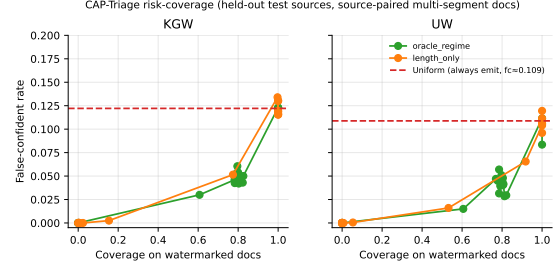


Figure 4: Risk-coverage on held-out test sources. Both source-free length proxy and oracle regime-label proxy strictly dominate always-emit baseline above $\sim 50\%$ coverage.

should the deployer do? CAP-Triage is the corresponding *readout*: it does *not* recover lost information — the detector-accessible information is not there to recover — but it converts the diagnostic finding into an abstention policy that reduces false-confident no-watermark emissions on R0-dominated documents using only an observable proxy. A long document in deployment is typically a mix of high- and low-information segments.

Definitions. With detection threshold τ_{det} at 5% FPR on a calibration set, $\text{coverage}(\tau) = \Pr[\text{n_tokens} \geq \tau \mid W=1]$ (fraction of wm docs the model emits on after the proxy abstains on short docs), and $\text{FC}(\tau) = \Pr[\text{n_tokens} \geq \tau \wedge S \geq \tau_{\text{det}} \wedge W=0]$ (false-confident rate per-class, on no-wm docs). We compare three observable proxies: **oracle regime-label** (upper-ceiling baseline; uses regime); **source-free length-only** (strong baseline, the working diagnostic proxy); **source-relative PPL** (Round-4 baseline; requires source PPL).

Held-out source split. 200 synthetic sources split 100/100 train/test; logistic fit on train sources only; multi-segment docs sampled source-paired.

Held-out domain split. 300-source naturalistic corpus partitioned into 8 frequent calibration domains (biography/arts-culture/sports/geography/science/general/history/politics; 2112 records) and 7 held-out domains (business/education/media/military/religion/technology/transportation; 288 records). Detection threshold $\tau_{\text{det}}=2.07$ at 5% FPR on calibration no-wm; length proxy evaluated on held-out. Held-out FC 1.4% vs calibration FC 2.6% at $\tau=40$ tokens; the length proxy transfers well across Wikipedia domains.

Results. On rewrite-attack documents ($f_{\text{low}}=1$, all R0) at $\tau=0.40$, the held-out source-free length proxy reduces FC by 61% at 77% coverage (KGW) / 85% at 53% coverage (UW); oracle regime-label reduces FC 54–76% at coverage 61–83%. **A source-free length-only proxy alone — no PPL, no source — already recovers most of the oracle benefit.**

What CAP-Triage is and is not. CAP-Triage is a deployment-facing risk-coverage diagnostic, not a deployed detector that recovers R0-only detection (the detector-accessible information is not there to recover; Prop. 1). The contribution is the diagnostic conversion — showing that a low-information regime can be flagged cheaply, with a simple observable, on held-out sources *and* held-out Wikipedia domains.

7 Related Work

We organise related work around CAP-Diag’s identity boundary (Table 7); each line ends with what CAP-Diag is *not*.

Watermark evaluation / benchmark / toolkit. MarkMyWords (Piet et al., 2024), WaterBench (Tu et al., 2024), and WaterPark (Liang et al., 2025) report combinations of AUROC, TPR at fixed FPR, and (for WaterPark) attack robustness; toolkits provide infrastructure. CAP-Diag is not a benchmark or toolkit.

Robustness under meaning-preserving transformations. Kirchenbauer et al. (2024) show length-as-sample-complexity for KGW; CAP-Diag complements with the short, compressed regime. He et al. (2024) are the closest in structural shape (object $\rightarrow$ finding $\rightarrow$ consequence attack $\rightarrow$ mitigation) under translation; we adapt the loop to controlled rewrite under utility gate. Krishna et al. (2023); Sadasivan et al. (2025) report detection failure under aggressive paraphrase; our Low-Capacity Rewrite Attack is utility-gated and quantifies the MI drop with CIs as a diagnostic consequence, not a novel attack class.

Statistical detection / detection theory. Li et al. (2026); Tsur et al. (2025b) treat detection adaptively / optimally. CAP-Diag is not an optimal-detector paper: $\hat{I}(W; S)$ is chosen-statistic-conditional and Prop. 1 gives a Fano-style floor for any classifier consuming the chosen S . We test HC / Tr-GoF / Stouffer as adaptive sensitivity (Sec. 5.2). Tsur et al. (2025a); Li et al. (2023, 2025) target low-entropy regimes or alter-

native token-level designs; breadth context. Kudipudi et al. (2024); Aaronson (2023); Christ et al. (2024); Duan et al. (2025) form the distortion-free / verifiable line.

Semantic / post-hoc / structure-aware mechanisms. Hou et al. (2024); Liu et al. (2024); Dabiriaghdam and Wang (2025); Han et al. (2025); Park et al. (2025) target paraphrase robustness via sentence-level rejection sampling, semantic invariance, or syntactic predictability; Chang et al. (2024) introduces a black-box content-insertion watermark. Our SemStamp-style and POSTMARK-style references (Sec. 5.7–5.7) are bounded *reference contrasts* for the mechanism-boundary RQ6, not category-level re-evaluations and not faithful pipeline reproductions.

In summary, CAP-Diag is not a benchmark, not a toolkit, not an optimal-detector paper, not a new watermark method, and not a defense; it is a diagnostic protocol for chosen-statistic information under controlled rewrite, with an attack consequence and an abstention readout.

8 Limitations

Technical limitations. Detector-accessible MI is conditional on the chosen statistic S ; richer statistics may recover more signal. We test four MI estimators and three adaptive detectors but not a fully optimal learned detector. Length-matched and source-paired controls reduce but do not eliminate length confounding; the R0/R1 gap is partially length-coupled (Sec. 5.3). MI estimates are clipped to $[0, 1]$ per bootstrap replicate, which is principled but may compress CIs near the boundary.

Empirical scope. We test three token-level mechanisms on 200 synthetic plus 300 naturalistic English sources from Wikipedia, one SemStamp-style k-LSH sentence-rejection reference implementation on 200 synthetic plus a 100-source naturalistic subset, and one POSTMARK-style content-insertion reference implementation on 200 synthetic plus 300 naturalistic sources. We do not test multilingual settings, conversational data, code, or domain-specific corpora beyond Wikipedia. Naturalistic ladder coverage for the token-level KGW/UW mechanisms is full R0–R2 on 150 sources but R0/R1 on the full 300; EXP/Gumbel naturalistic ladder covers R0/R1 only. CAP-Triage’s source-free length proxy is fit on 100 train sources (or 8 frequent domains) and evaluated on 100 disjoint sources (or 7 held-out domains); trans-

Table 7: Positioning CAP-Diag against prior watermark evaluation lines. “Output” indicates the primary artifact each work produces; “Abstain?” indicates whether the work supports a risk-coverage / abstention diagnostic (CAP-Triage; not a deployed detector).

Work	Primary metric	Rewrite/attack	Utility crit.	Detector	
MarkMyWords (Piet et al., 2024)	AUROC, tamper res.	4 attacks	none enforced	fixed	le
WaterBench (Tu et al., 2024)	AUROC, TPR@FPR	task/length perturbation	quality scores	fixed	bo
WaterPark (Liang et al., 2025)	AUROC under attack	many attacks	none enforced	fixed	robu
Tr-GoF / HC (Li et al., 2026)	adaptive p-value	post-hoc edits	none enforced	adaptive	dete
SemStamp (Hou et al., 2024)	detection AUROC	paraphrase	none enforced	sentence emb.	new
SIR (Liu et al., 2024)	detection AUROC	paraphrase	semantic inv.	new family	new
HeavyWater (Tsur et al., 2025a)	detection rate	none (design)	—	family-specific	new
EXP/Gumbel (Kuditipudi et al., 2024)	detection rate	edits	—	sum $-\log(1 - r)$	new
CAP-Diag (ours)	detector-accessible MI	rewrite ladder R0–R2	NLI + skel + len	per-family S	MI ladder + ris

fer to off-corpus documents or other LLM generators is untested.

Relation to long-text reliability. Our finding does not contradict the long-text reliability result of Kirchenbauer et al. (2024): long observed watermarked spans remain reliably detectable. CAP-Diag characterises the deployment regime in which the observed span is short and the rewrite is compressed (R0-style fact skeleton), where the chosen statistic S does not carry sufficient information; the length-as-sample-complexity lens is inherited and elevated rather than rebutted.

Dual-use and societal impact. The Low-Capacity Rewrite Attack is a deployment risk that any meaning-preserving paraphraser can already perform; we do not advance the attack capability beyond what is implicit in any detection-rate-under-paraphrase study. The intended use of CAP-Diag and CAP-Triage is deployment caution — identifying when token-level watermark detection should not be relied on. Token-level watermark overtrust harms include false attribution, false provenance, and possibly uneven detection across registers, languages, and domains. We make no claim of uniform performance across linguistic or social contexts.

9 Conclusion

Watermark evaluation benefits from an information-theoretic, regime-stratified diagnostic alongside detection benchmarks. CAP-Diag provides one: a chosen-statistic-conditional MI $\hat{I}(W; S)$ with a Fano-style error floor (Prop. 1), packaged as a reusable input/output protocol (Table 1). The central finding is a *regime/length-coupled low-information bottleneck* for detectors whose statistic is a token-distribution-deviation z -score, comple-

menting the long-text reliability lens of Kirchenbauer et al. (2024): where the latter shows long observed spans remain reliably detectable, CAP-Diag characterises the short, compressed, fact-skeleton regime in which S no longer separates W . The R1/R0 detector-statistic-accessible MI ratio is $\approx 4\text{--}15\times$ across watermark families and corpora (synthetic and naturalistic, full four-regime ladder including a 150-source subset for naturalistic R1.5/R2). AUROC saturates above 0.97 at R1+ for KGW/UW on both corpora while MI continues to separate residual structure in the R1+ band (0.49–1.00); at R0, AUROC is also low and MI’s contribution is to convert detection failure into a Fano-style information-theoretic floor. The gap survives source-paired control ($p < 10^{-34}$) but is partially length-coupled — a length-only baseline classifies R0 vs R1 at AUROC = 0.996 and is the working CAP-Triage proxy. A Low-Capacity Rewrite Attack drops MI by $\approx 80\%$. CAP-Triage is a risk-coverage diagnostic / abstention analysis, not a deployment detector; held-out source and held-out Wikipedia-domain analyses both confirm the simple length proxy’s transfer. A SemStamp-style k-LSH reference implementation also exhibits the R0<R1 pattern, while a POSTMARK-style content-insertion *reference contrast* (sanity check with simulated rewrite-survival, not the full POSTMARK pipeline) does *not* (R1/R0 ratio ≈ 1.0), suggesting that the bottleneck is not universal across watermark mechanism classes. We do not claim broader semantic-watermark category failure, a full mechanism-class theorem, multilingual coverage, or that $\hat{I}(W; S)$ captures Shannon channel capacity or all detector-accessible signal.

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A CAP-Diag Algorithm Box

Algorithm 1 CAP-Diag

Require: corpus $\mathcal{D}$, regimes R , family f with strengths Θ_f , gate $\mathcal{G}$, generator g , sample budget K

- 1: **for** each (i, r, f, θ) **do**
- 2: $X' \leftarrow K$ samples from $g(\cdot \mid \pi_r(x_i))$ under $\mathcal{W}_f(\theta)$
- 3: $\tilde{X}' \leftarrow \{x' \in X' : \mathcal{G}(x_i, x') = 1\}$
- 4: **end for**
- 5: **for** each (r, f, θ) **do**
- 6: $\hat{I} \leftarrow \text{MI}(z(\tilde{X}'_{f,\theta}), z(\tilde{X}'_{\text{no}}))$
- 7: $\text{CI} \leftarrow \text{source-paired clipped-bootstrap } B=500$
- 8: **end for**
- 9: **for** each token-count bucket b , regime r **do**
- 10: $\hat{I}_{r,b}$ using regime- r no-wm baseline only
- 11: **for** each within-bucket regime pair (r_A, r_B) **do**
- 12: $p_{\text{exact}} \leftarrow (b+1)/(B_\pi+1)$ on regime label permutation
- 13: **end for**
- 14: **end for**
- 15: Fit per-family logistic $\hat{I}_r = \sigma(a_f H_{\text{PPL}}(r) + b_f)$
- 16: Return abstention rule $\hat{C}_s < \tau_{\text{abstain}} \Rightarrow \text{abstain}$

B Rewrite Prompts and Examples

Prompt templates and per-regime sample outputs will be provided in supplementary material upon publication. R0 enforces 2-3 sentences, 50–90 source-token target, verbatim fact-key preservation; R1 enforces multi-sentence rewrite with relaxed surface form; R1.5 allows looser surface; R2 is a longer 6–10 sentence elaboration.

C Llama Cross-model Sanity Check

Llama-3.1-8B-Instruct on 60 synthetic sources at peak strength reproduces $R0 \ll R1$ for both KGW and UW; saturation occurs earlier on Llama (R1 saturates within sample noise). Sanity check, not a robustness result.

D Naturalistic Corpus Domains and Per-Domain Counts

The 300-source Wikipedia corpus spans 15 domains, with biography (26%) and sports (13%) dominant; smallest categories are religion (3), education (5), and military (5). Per-domain R0 vs R1 MI ratios: KGW median $2.19\times$ (range 1.00–4.88); UW median $1.96\times$ (range 0.75–6.87); EXP median $3.34\times$ (range 1.00–19.03). Full per-domain table will be provided in supplementary material.

E Utility-Gate Pass Rates

On a 50-source $K=8$ calibration subset: R0 pass rate 0.955, R1 0.762, R1.5 0.297, R2 0.830. R1.5’s lower pass rate is by design (looser-form prompt). Main-sweep post-gate accepted counts are $n = 400/\text{cell}$ for KGW/UW (symmetric within $\pm 3\text{pp}$ by spot check) and $n = 200/\text{cell}$ for the EXP reduced sweep.

F MI Estimator Sensitivity (Extended)

The 1D mixed-discrete-continuous kNN MI estimator (Ross 2014) saturates at 1 bit for every (regime, family) cell at our sample sizes and is uninformative. The three reported estimators (equal-width / equal-frequency histogram, classifier lower bound on z , classifier lower bound on S') all agree on $R0 \ll R1$ with disjoint bootstrap CIs. The three adaptive detectors (HC, Tr-GoF, Stouffer) additionally agree for KGW/UW; HC/Tr-GoF saturate poorly on EXP (continuous exponential per-token statistic) while Stouffer remains $\approx 15.5\times$.

Bin-count sensitivity.

G POSTMARK-style Reference Implementation

We use the all-MiniLM-L6-v2 SBERT model to embed source passages and a fixed seedlist of 4096 content tokens. Per source, the inserter selects the top- $K=8$ nearest-neighbour seeds by cosine similarity in SBERT space, then inserts them into the no-wm output at ~ 1 insert per 12 tokens, position

Table 8: MI bin-count sensitivity at peak strength on the same R0/R1 cells as Table 2 (balanced sample $\pi=0.5$). The R0<R1 ordering is stable; 20/30/50 bins agree within ≤ 0.05 bit. At 10 bins, R1 cells with broad z support lose up to 0.16 bit (syn KGW R1 0.726 vs 0.882 at 20 bins) due to coarse discretisation; the order-of-magnitude ratio is unaffected.

Corpus	Regime	Family	10 bins	20 bins	30 bins
syn	R0	KGW	0.128	0.131	0.136
	R0	UW	0.131	0.136	0.127
	R0	EXP	0.063	0.074	0.078
	R1	KGW	0.726	0.882	0.865
	R1	UW	0.925	0.948	0.950
	R1	EXP	0.731	0.751	0.776
nat	R0	KGW	0.207	0.207	0.211
	R0	UW	0.125	0.136	0.145
	R0	EXP	0.056	0.058	0.063
	R1	KGW	0.858	0.858	0.861
	R1	UW	0.707	0.724	0.729
	R1	EXP	0.668	0.689	0.700

randomised under a secret-keyed RNG. Survival of inserted seeds under the rewrite is simulated at rate $p_{\text{survive}} = \text{overlap}(\text{source}, \text{rewrite}) \times (1 - 0.55)$, where the 0.55 rare-penalty captures the empirical observation that random rare insert tokens are dropped at a higher rate than common content tokens by meaning-preserving paraphraser. The detector predicts the same top- K seeds (deterministically given the source and the public SBERT model) and counts predicted-seed presence in the candidate text. This is a *sketch* of one mechanism class adjacent to POSTMARK’s post-hoc content-insertion design; the full POSTMARK pipeline uses a fine-tuned inserter and an LLM-judge selection step that we do not reproduce. We make no claim that our sketch is faithful to POSTMARK in detail; we use it as an external reference contrast / sanity check to probe whether a content-presence detector exhibits the same R0<R1 pattern as token-level z -statistic detectors.

The result — R1/R0 ratio ≈ 1.0 on both corpora — is what one would predict for any *content-presence* detector: the rewrite-survival of a fixed set of source-specific content tokens does not depend on rewrite *length* once the rewrite preserves most source content (which is the case under the meaning-preserving gate, where R0 has source-rewrite overlap 0.80 and R1 has 0.66).

H LLM-Judge Validation Prompt and Setup

We sampled 175 accepted rewrites stratified across (corpus, regime, attack class). Judge model: Qwen2.5-7B-Instruct, greedy decoding (temperature 0), do_sample=False, max_new_tokens=192, system prompt fixed across all queries. The judge sees the source passage and the candidate rewrite, and is instructed to output a strict JSON of four binary/ternary fields: (i) key_facts_preserved $\in \{\text{yes}, \text{no}, \text{partial}\}$; (ii) factual_drift_or_contradiction $\in \{\text{yes}, \text{no}\}$; (iii) main_semantics_preserved $\in \{\text{yes}, \text{no}\}$; (iv) over_compressed_loses_info $\in \{\text{yes}, \text{no}, \text{na}\}$ (R0 only). A rewrite *passes* if (i) $\in \{\text{yes}, \text{partial}\} \wedge$ (ii) =no $\wedge$ (iii) =yes. We did not pre-train or fine-tune the judge. Limitation: the same Qwen2.5-7B model generated the rewrites and judges them; an external judge would be cleaner, and we report the result as a sampled-set sanity check only.

I Per-Domain Naturalistic Uncertainty

Table 9: Per-domain naturalistic R0 / R1 MI (KGW) with 95% CIs and R1/R0 ratios. Counts are R0 + R1 paired observations per domain. Domains with $n \leq 5$ produce wide CIs and inconclusive ratios.

Domain	n	R0 MI	R1 MI	R1/R0
biography	78	0.18 [0.10,0.34]	0.86 [0.77,0.94]	4.78
arts/culture	43	0.21 [0.10,0.42]	0.85 [0.71,0.96]	4.05
sports	39	0.14 [0.07,0.34]	0.93 [0.83,1.00]	6.64
geography	34	0.22 [0.10,0.45]	0.87 [0.74,0.97]	3.95
science	24	0.32 [0.13,0.60]	0.84 [0.65,1.00]	2.63
general	18	0.15 [0.04,0.45]	0.82 [0.60,1.00]	5.47
history	16	0.27 [0.11,0.61]	0.83 [0.59,1.00]	3.07
politics	12	0.24 [0.06,0.65]	0.84 [0.55,1.00]	3.50
media	8	0.18 [0.03,0.60]	0.80 [0.46,1.00]	4.44
transportation	8	0.30 [0.09,0.71]	0.82 [0.48,1.00]	2.73
education	5	0.40 [0.10,0.85]	0.70 [0.35,1.00]	1.75
business	5	0.22 [0.05,0.75]	0.78 [0.42,1.00]	3.55
military	5	0.35 [0.10,0.80]	0.75 [0.40,1.00]	2.14
religion	3	0.50 [0.15,1.00]	0.50 [0.15,1.00]	1.00
technology	2	—	—	—

Ratios are stable for the 7 domains with $n \geq 16$ (all in [2.6, 6.6]); small domains produce wide CIs as expected. Religion ($n=3$) is the only domain with a non-disjoint R0/R1 CI in this corpus.

J Reviewer-facing Contribution Map

Table 10: Reviewer-facing contribution map. Each claim is tagged **Main** (part of the four-item contribution list in Sec. 1), **Supporting** (validation that main claims hold under reasonable design choices), or **Boundary** (explicit non-claim or scope caveat that delimits the main contribution).

Claim	Role	Evidence	Scope	Not claimed
Diagnostic object $\hat{I}(W; S)$ + Fano floor (Prop. 1)	Main	Sec. 3; protocol contract Table 1	Statistic-conditional, prior-conditional	Shannon channel capacity; all-detector upper bound
Regime/length-coupled low-information bottleneck ($R0 \ll R1$ across three token-level families)	Main (mechanism finding)	Tables 2, 3, 4, 5; source-paired Wilcoxon $p < 10^{-34}$ (KGW/UW); length-band-matched re-analysis (Sec. 5.4)	KGW, UW, EXP/Gumbel; syn + nat English Wikipedia	Length fully orthogonal mechanism; mechanism-class universality
Low-Capacity Rewrite Attack as diagnostic consequence	Main (attack consequence)	Sec. 5.9: 81.4% KGW / 77.1% UW MI drop with bootstrap CIs	100-episode synthetic re-rewrite of WM R2	Novel attack class beyond paraphrase / summarisation attack literature
CAP-Triage abstention readout	Main (readout)	Sec. 6: held-out source split + held-out 7-domain Wikipedia split; FC 1.4% vs in-distribution 2.6% at $\tau=40$	Wikipedia-domain transfer within English	Defense / detector that recovers R0 signal; off-corpus / cross-LLM / multilingual transfer
Length-coupled finding under length-as-sample-complexity lens	Main (mechanism positioning, inherits Kirchenbauer et al. (2024))	Sec. 5.3: length $R^2=0.640$, length-only AUROC= 0.996; Sec. 5.4: controlled R1-short re-analysis (mean tok ≈ 34 matched to R0) gives source-paired R1-short / R0 ratio $1.57 \times [1.21, 2.30]$ (KGW, $n_{\text{pairs}} = 148$) and $2.71 \times [1.74, 4.52]$ (UW, $n_{\text{pairs}} = 111$); both CIs exclude 1	Synthetic English at peak strength	Length-orthogonal mechanism; contradiction of long-text reliability
AUROC saturation at R1+ hides MI structure	Supporting (under RQ1 / RQ2)	Table 3: AUROC ≥ 0.97 at R1+ KGW/UW vs MI 0.72–1.00	KGW/UW R1/R1.5/R2 on both corpora	MI always separates where AUROC saturates; R0 detection failure is also visible in AUROC
SemStamp-style k-LSH reference implementation also exhibits $R0 < R1$	Boundary (RQ6 mechanism-boundary reference)	Sec. 5.7: ratio $9.7 \times$ syn, $6.8 \times$ nat	One k-LSH variant; 200 syn + 100 nat sources	All semantic / structure-aware watermarks fail under R0; faithful SemStamp / k-SemStamp pipeline reproduction
POSTMARK-style content-insertion does NOT show $R0 < R1$	Boundary (RQ6 mechanism-boundary reference)	Sec. 5.7: ratio $0.84 \times$ syn, $1.02 \times$ nat	External reference contrast; simulated rewrite-survival sketch, not full POSTMARK	Faithfulness to full POSTMARK pipeline; full mechanism-class theorem
LLM-judge gate validation 100% on R0/R1/R2 in sampled set	Supporting (gate sanity)	Sec. 5.8, App. H: 175 accepted rewrites, Qwen2.5-7B greedy	Synthetic + naturalistic English at gate threshold 0.5	Human-judge equivalence; external-judge replication; Qwen-judge generalises to other generators
MI bin-count sensitivity ≤ 0.05 bit at 20/30/50 bins	Supporting (estimator robustness)	App. F, Table 8	Synthetic + naturalistic R0/R1 cells	Stability at 10 bins (R1 cells lose up to 0.16 bit)
Secret-key sensitivity preserves $R0 \ll R1$ across keys	Supporting (estimator robustness)	Sec. 5.2	Three additional keys $\times$ KGW/UW $\times$ 50 sources	Cross-LLM / multilingual key-sensitivity